

Sample Paper 2 — Answer Key

Total: 60 marks (45×1 mark + 5×3 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **C) Potato** — Apple, Banana, and Mango are fruits; Potato is a vegetable.
2. **B) Sky** — A fish's home is water; a bird's natural space is the sky.
3. **A) EPH** — The rule is "+1 letter": $D \rightarrow E$, $O \rightarrow P$, $G \rightarrow H$.
4. **C) 160** — Each term doubles: 5, 10, 20, 40, 80, 160.
5. **B) 17th** — Position from the right = $25 - 9 + 1 = 17$.
6. **B) South** — Two 90° clockwise turns from North = South.
7. **A) 3, 1, 4, 2** — Natural growth order: Infant $\rightarrow$ Child $\rightarrow$ Teenager $\rightarrow$ Adult.
8. **C) 39** — 31, 37, 41 are prime; $39 = 3 \times 13$ is not.
9. **B) Saturday** — 26 days = 3 full weeks (21 days, same day: Monday) + 5 more days $\rightarrow$ Tue, Wed, Thu, Fri, Sat.
10. **B) Son** — "My father's only daughter" is Meena herself (since she has no sisters); her only son is her own son.
11. **C) 63** — The pattern is "double and add 1": $3 \times 2 + 1 = 7$, $7 \times 2 + 1 = 15$, $15 \times 2 + 1 = 31$, $31 \times 2 + 1 = 63$.
12. **A) OQWUG** — The rule is "+2 letters": $M \rightarrow O$, $O \rightarrow Q$, $U \rightarrow W$, $S \rightarrow U$, $E \rightarrow G$.
13. **C) Triangle** — Cone, Cylinder, and Sphere are 3-D solids; a Triangle is a flat (2-D) shape.
14. **D) S** — $S > P > Q > R$, so S is the oldest.
15. **C) 37** — The gaps increase by 2 each time (+3,+5,+7,+9,+11): 2,5,10,17,26,37.

Section 2 — Mathematical Reasoning

16. **A) 91,356** — $56,789 + 34,567 = 91,356$.
17. **A) 44,325** — $90,003 - 45,678 = 44,325$.
18. **A) 41,103** — $4,567 \times 9 = 41,103$.
19. **A) 10,864** — $86,912 \div 8 = 10,864$.
20. **C) 16** — HCF of 32 (2^5) and 48 ($2^4 \times 3$) is $2^4 = 16$.
21. **A) 36** — Multiples of 9: 9,18,27,36... Multiples of 12: 12,24,36... First common one = 36.
22. **A) 8/10** — Multiply top and bottom of $4/5$ by 2.
23. **A) 3/4** — Same denominator: $3+2+1=6$, so $6/8$, which simplifies to $3/4$.
24. **A) 9/20** — $0.45 = 45/100$, which simplifies ($\div 5$) to $9/20$.

25. **C) 84** — $35\% \text{ of } 240 = 0.35 \times 240 = 84$.
26. **B) 70%** — $7/10 = 0.7 = 70\%$.
27. **C) 30** — 1 part = $18 \div 3 = 6$; larger = 5 parts = 30.
28. **B) 25%** — Profit = $400 - 320 = ₹80$; profit % = $80/320 \times 100 = 25\%$.
29. **C) ₹1,200** — SI = $(4000 \times 6 \times 5) \div 100 = ₹1,200$.
30. **A) 5,080 g** — 5 kg = 5,000 g; + 80 g = 5,080 g.
31. **C) 135 min** — $2 \times 60 + 15 = 135$ minutes.
32. **B) 60 cm** — Perimeter = $2 \times (19 + 11) = 60$ cm.
33. **B) 196 sq cm** — Area = $14 \times 14 = 196$ sq cm.
34. **B) 1** — A (non-equilateral) isosceles triangle has exactly 1 line of symmetry.
35. **B) 3** — Three-quarters of $360^\circ = 270^\circ$; $270 \div 90 = 3$.

Section 3 — Everyday Mathematics

36. **B) 80,000** — $3,200 \times 25 = 80,000$.
37. **A) ₹3,775** — $8000 - 2350 - 1875 = 3,775$.
38. **B) 180 L** — $2/3$ of 540 = 360 L used; $540 - 360 = 180$ L left.
39. **C) ₹50** — $900 \div 18 = 50$.
40. **B) 324** — $9 \times 36 = 324$.
41. **C) ₹840** — SI = $(3000 \times 7 \times 4) \div 100 = ₹840$.
42. **B) 15%** — Profit = $2070 - 1800 = ₹270$; profit % = $270/1800 \times 100 = 15\%$.
43. **C) 125 km** — 1 h 40 min = $5/3$ hours; $75 \times 5/3 = 125$ km.
44. **B) 4 cups** — $2/3 \times 6 = 4$ cups.
45. **C) 108** — $45\% \text{ of } 240 = 0.45 \times 240 = 108$.

Section 4 — Achievers Section (3 marks each)

46. **C) 300** — If the original number is x : $0.75x = 225 \rightarrow x = 300$.
47. **B) 35** — Three consecutive multiples of 5: $n, n+5, n+10$, summing to $3n+15 = 90 \rightarrow n = 25$; the largest is $25+10 = 35$.
48. **C) ₹10,000** — SI = $(P \times 8 \times 4) \div 100 = 0.32 \times P = 3,200 \rightarrow P = 3,200 \div 0.32 = ₹10,000$.
49. **C) ₹1,800** — After the dress, $2/3$ of the money remains. After shoes ($1/4$ of that spent), $3/4$ of $2/3 = 1/2$ of the original remains. So $1/2$ of the total = ₹900 $\rightarrow$ total = ₹1,800.
50. **B) 15** — Let the ages be $2x$ and $3x$. After 5 years: $(2x+5)/(3x+5) = 3/4 \rightarrow 4(2x+5) = 3(3x+5) \rightarrow 8x+20 = 9x+15 \rightarrow x = 5$. The elder's present age = $3 \times 5 = 15$.
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Scoring guide (out of 60): 42+ is Level-2 cutoff range — strong. 30–41 is solid Level-1 performance. Below 30 — revisit the relevant topics before attempting Sample Paper 3.